

VALGA, Estonia

WMO: 262470

Lat: 57.7900N

Lon: 26.0378E

Elev: 215

StdP: 14.58

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.3	-1.0	-12.2	2.9	-7.9	-5.3	4.2	0.1	14.9	29.8	14.0	30.1	1.9	60	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	82.5	67.2	79.3	65.4	76.2	64.0	69.5	78.8	67.4	75.8	65.6	73.7	5.7	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	97.3	73.8	64.3	90.8	71.8	62.4	84.9	69.6	33.6	79.1	32.0	75.9	30.5	73.5	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.2	12.5	11.2	DB	-13.1	86.9	8.3	3.0	-19.1	89.0	-23.9	90.8	-28.6	92.5	-34.7	94.6	(4)
(5)			WB	-13.5	72.2	8.2	2.4	-19.3	73.9	-24.1	75.3	-28.7	76.6	-34.6	78.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.1	23.3	23.2	30.7	42.7	52.2	59.4	64.4	61.7	53.1	42.9	34.3	27.9		
	DBStd	16.58	11.51	11.93	8.23	7.53	7.72	5.99	5.56	5.29	5.99	7.23	8.15	10.15		
	HDD50	3912	828	750	598	242	63	3	0	0	37	236	470	685		
	HDD65	8107	1293	1170	1063	669	402	186	79	132	358	685	920	1150		
	CDD50	1398	0	0	0	23	132	286	447	363	131	16	0	0		
	CDD65	116	0	0	0	0	6	19	61	29	1	0	0	0		
	CDH74	1150	0	0	0	3	83	214	566	274	10	0	0	0		
	CDH80	211	0	0	0	0	10	33	125	43	0	0	0	0		
	WSAvg	5.1	5.6	5.5	5.7	5.4	4.8	4.8	4.3	4.2	4.4	5.2	5.5	6.1		
	PrecAvg	26.4	2.0	1.6	1.5	1.4	2.2	3.1	2.7	3.0	2.0	2.6	2.2	1.9		
(15) Precipitation	PrecMax	32.7	3.9	2.8	3.3	3.9	4.4	6.3	5.7	5.7	3.9	4.6	4.1	3.5		
	PrecMin	19.3	0.6	0.4	0.5	0.3	0.6	1.3	0.7	0.0	0.4	0.2	0.4	0.7		
	PrecStd	3.7	0.9	0.7	0.7	0.8	0.9	1.2	1.3	1.6	0.9	1.1	0.9	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.6	43.2	56.9	72.9	81.3	83.7	87.3	84.5	75.3	62.6	51.2	48.2		
		MCWB	43.6	40.1	44.6	55.6	64.7	67.0	70.3	68.3	62.2	55.9	48.4	46.4		
	2%	DB	40.2	40.3	49.2	67.0	75.9	79.8	83.5	80.2	69.8	57.9	48.5	44.4		
		MCWB	38.7	38.2	41.6	52.6	60.7	64.3	68.6	66.5	60.1	54.2	46.4	42.8		
	5%	DB	37.7	38.4	44.3	61.9	71.8	76.0	80.3	76.9	66.3	55.3	46.7	41.9		
		MCWB	36.4	36.8	38.9	49.7	57.8	63.4	66.8	64.7	58.6	52.4	45.1	40.6		
	10%	DB	35.9	36.7	41.0	57.2	67.4	72.0	76.8	73.5	63.3	53.1	44.8	39.5		
		MCWB	34.8	35.3	37.1	47.2	55.7	60.6	65.4	63.2	57.1	50.7	43.5	38.3		
	0.4%	WB	43.4	40.9	46.4	57.6	67.4	69.8	72.8	71.4	64.0	57.2	49.2	46.5		
		MCDB	44.2	42.2	55.2	70.0	78.1	79.6	83.8	80.6	70.4	60.0	50.6	48.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.0	38.6	42.3	53.9	63.1	66.9	70.3	68.3	61.7	54.8	46.9	43.0		
		MCDB	40.0	39.9	47.3	64.0	71.8	76.3	80.4	76.1	67.6	57.2	48.0	43.9		
	5%	WB	36.6	37.0	39.7	51.1	60.0	64.5	68.4	66.1	59.8	53.0	45.3	40.7		
		MCDB	37.4	38.1	43.3	59.8	68.2	72.9	77.0	74.2	64.7	55.1	46.5	41.9		
	10%	WB	34.9	35.4	37.8	48.3	57.1	62.2	66.5	64.2	58.0	51.0	43.5	38.3		
		MCDB	35.7	36.6	40.4	56.1	65.9	69.6	74.5	71.8	62.2	52.8	44.7	39.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	10.5	13.3	18.1	20.0	17.8	17.9	17.9	15.6	11.0	7.1	7.0		
		MCDBR	6.3	8.1	18.0	27.5	26.3	23.9	23.2	24.1	20.4	13.3	8.4	6.6		
		MCWBR	6.0	6.9	11.7	14.5	13.0	11.1	10.3	11.4	11.7	9.6	7.4	6.5		
		MCDBR	6.3	7.1	14.8	23.7	21.8	20.1	20.8	20.6	17.4	12.3	7.8	6.5		
	5% WB	MCWBR	6.2	6.5	10.5	13.1	12.0	10.6	10.1	10.6	10.9	9.4	7.2	6.6		
(40) Clear-Sky Solar Irradiance	taub		0.277	0.307	0.338	0.374	0.368	0.368	0.394	0.390	0.359	0.336	0.317	0.272		
	taud		2.139	2.144	2.192	2.296	2.377	2.406	2.350	2.360	2.430	2.402	2.236	2.139		
	Ebn at Noon		178	226	254	265	275	276	265	257	248	219	161	141		
	Edh at Noon		16	26	33	35	35	34	36	33	26	21	16	12		
(44) All-Sky Solar Radiation	RadAvg		129	349	797	1256	1687	1747	1670	1325	851	387	126	79		
	RadStd		27	64	105	144	173	136	161	131	95	57	18	13		

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	+0.77	N/A	N/A	N/A	N/A	-284	N/A	N/A	N/A	(46)
(47)	Regional (34 neighbors)	+0.70	N/A	N/A	N/A	N/A	-256	N/A	N/A	N/A	(47)

Nomenclature: See separate page